

Hemaang Sood

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Skills

Languages

Python, JavaScript, Go, Java, Kotlin, R, UML, LaTeX (Qmd, Rmd).

Web Development

Django, Spring boot, React.js (NextJS, Remix), FastAPI, Next-auth, Redux, Tailwind CSS, NodeJS, Three.js, React Three Fiber (R3F).

Cloud, DevOps & Tools

AWS (EC2, S3), Git, GitHub, Docker, Nginx, Postman.

CI/CD & Testing

GitLab CI/CD, Docker-in-Docker, pytest, Poetry, PostGIS.

Game Dev

Unity3D, Pygame

AI & ML

PyTorch, TensorFlow, OpenCV, OpenAI Gym, OpenAI API, HuggingFace(Transformers), xgboost.

Data Analysis

Pandas, Numpy, Matplotlib, Plotly, Tableau, SQL.

Automation & Scraping

Selenium, BeautifulSoup, Web-Scraping

Databases

MongoDB (Atlas), PostgreSQL (Supabase), Supabase, FAISS, Qdrant, Redis.

Experience

Synechron

Jr Associate

- Completed introductory sessions on **Capital Markets** and **Commodities**, developing foundational understanding of financial systems and trading concepts.
- Built a full-stack project for an **Amusement Park Management System** using **MERN** with **Vite.js** and **Tailwind CSS**, focusing on modular UI design, API integration, and database connectivity.
- Built a backend server project for an **Insurance Management System** using Springboot, focusing scalability security and modularity.

Bharat Electronics Limited (BEL)

Intern

- Developed an Authentication and Authorization module for DRDO as part of the internship as part of a secure intranet website.
- Worked on a Large Language Model (LLM) interface to locally process and extract insights from sensitive documents using **RAG**. Integrated a **voice input pipeline** using Vosk to transcribe user speech into queries, enhancing accessibility and usability.

Education

University of Waterloo

Electrical & Computer Engineering, MEng

GPA - 3.95 / 4.0

2026 – Present

Certifications

- AWS Cloud Practitioner Essentials.
- Reinforcement Learning.
- Django Rest API.

Conference Publications & Presentations

- Hemaang Sood et al., "Improving Face Recognition for Smart Home Security: A Study of ResNet, Siamese Networks, and Attention Mechanisms" Presented at IEEE COMPSIF 2025.

Personal Projects

Routed: Travel Partner Recommendation System Group

- Designed and developed a microservices-based platform to match travelers with compatible trip partners using trip overlap, preferences, and geospatial proximity.
- Implemented a FastAPI backend with JWT authentication, trip management APIs, and a matching system powered by PostgreSQL and PostGIS for spatial queries.
- Built an event-driven notification pipeline using Kafka and a Go-based worker service that processes match events and stores notifications in MongoDB.
- Built an Kotlin microservice with coroutines to extract semantic embeddings from bios.
- **Technologies Used:** FastAPI, React, TypeScript, PostgreSQL, PostGIS, Kafka, MongoDB, Docker, Go, SQLAlchemy.

Facial and Voice Recognition-Based Door Lock System: Group

- Achieved 98.7% accuracy in preventing unauthorized access by integrating facial and speech recognition algorithms for secure authentication.
- Processed and augmented the YaleFace dataset, utilizing a custom YOLO model to extract, align, and normalize facial features for improved model training.
- Implemented a Siamese Network to optimize feature comparison, enhancing the precision and robustness of facial recognition for reliable access control.
- Built Backend in Django and made a NGINX reverse proxy.
- Published paper in **IEEE Compsif 2025** for the same.
- **Technologies and Tools:** OpenCV, PyTorch, Google Speech API, RESTful APIs, Python, Raspberry Pi, Hybrid Neural Networks, Docker, React, PostgreSQL, Tailwind, Shad CN, AWS EC2, Django, NGINX.

Virtual POCT — AI-Powered Quality Control Assistant: Individual

- Built a full-stack AI-powered quality control platform for Point-of-Care Testing (POCT), enabling health-care operators to troubleshoot devices through a conversational assistant with auditable QC workflows.
- Designed a streaming WebSocket architecture using FastAPI and LangChain to deliver real-time LLM responses while supporting concurrent multi-session conversations.
- Implemented a deterministic quality control evaluation engine that combines operator input with database context to classify QC scenarios and automate compliance actions.
- Developed a Dockerized microservice architecture with React, FastAPI, PostgreSQL, and Nginx, including JWT authentication, role-based access control, and persistent session history.
- **Technologies Used:** FastAPI, React, TypeScript, PostgreSQL, SQLAlchemy, Docker, Nginx, WebSockets, LangChain, Langfuse, Google Gemini, JWT.

Personal Profile Website: Individual

- Built personal profile website.
- Optimized site for SEO and made it comply with standards by google search console.
- **Technologies Used:** ReactJS, Three.js, React Three Fiber, Tailwind.

Stock Analysis using Machine Learning: Individual

- Collected, cleaned, and transformed historical stock data for 25 stocks, integrating 10 global economic factors to enhance model performance and generate actionable insights.
- **Technologies and Techniques:** LSTM, GRU, CNN, Reinforcement Learning, Hybrid Neural Networks, PyTorch, Python, AWS (EC2, S3), Pattern Recognition, Data Preprocessing, Feature Engineering, Statistical Analysis.

ETF Volatility Forecasting with Regularized Regression: Individual

- Forecasted next-day volatility for **SPY, QQQ, GLD, IWM, and TLT** using 2010–2025 daily market data and engineered rolling volatility, EMA, return, volume, and statistical features.
- Compared **OLS, Ridge, LASSO, and regression tree** models using chronological train/test evaluation.
- Applied **LASSO feature selection** and extended the model with cross-asset volatility signals, achieving the strongest overall out-of-sample performance among the evaluated models.
- **Technologies Used:** R, tidyquant, glmnet, rpart, dplyr, ggplot2, Statistical Modeling, Feature Engineering

MERN-Stack Based Chat App: Individual

- Created Highly secure end-to-end encrypted chat app with 256 bit AES-256 encryption.
- **Technologies Used:** MongoDB, Express.js, React.js, Node.js, Tailwind, Next-Auth, Cryptography.

Resume Builder: Individual

- Built an AI-powered resume optimization tool that parses LaTeX resumes, tailors content to job descriptions, and generates polished, job-specific resume output.
- Agent utilizes multi-step process to parse optimize and output the the optimized resume.
- **Technologies Used:** Python, OpenAI API, LaTeX, SQLite, Jinja2, Rich, pdflatex, python-dotenv

Reinforcement Learning for Snake Game: Individual

- Created Convolutional Neural network that learnt to complete snake game within 45 minutes of training.
- **Technologies Used:** Python, TensorFlow, Deep Q-Learning, RL, OpenAI Gym, PyGame, CNN, NumPy, Matplotlib.

Hand Gesture-Controlled Omni-Directional Wheelchair: Individual

- Created lifesize wheelchair controlled over BLE with 2 primary(perpendicular) axis of motion.
- Built the same in **Unity3D** to test the gesture detection system.
- **Technologies Used:** Mediapipe, OpenCV, Arduino, BLE, Python.

Achievements

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- Received **funding from State government** initiative for final year project through KSCST.
 - Selected for **College Open Day Presentation** three times in a row (Sem 3, 4, 5) – Won **1st and 2nd place** at the department level.
 - Projects selected for **college-level tech fest** presentations in 2023 and 2024.
 - Received **Appreciation Letter** from Software SBU, BEL, for contributions to the DRDO project.
 - Received a cash prize from the Training and Placement Section for being one of the top performers in the HackerEarth Coding Assessment held at BMSIT&M.